Unit 4 Quantum computing

WHY QUANTUM COMPUTING IS NEEDED?

LIMITATIONS OF CLASSICAL COMPUTING

The limitations of classical computing primarily centre on its inability to efficiently handle certain types of complex, high-dimensional problems.

- **Inability to Exactly Model Nature/Chemistry**: Classical computers cannot fully represent the complexity of even small molecules. For instance, modelling a single caffeine molecule at a single instant would require approximately 10^{48} bits—a number comparable to 1% to 10% of the atoms on Earth. Because "Nature isn't classical," traditional storage cannot provide enough space to hold this much information for exact representation.
- **Exponential Growth of Resource Requirements**: For certain problems, as the size of the problem increases, the resources (time, memory, or storage) required by a classical computer grow exponentially. This rapid growth makes such problems "impossible" for classical systems to solve within a reasonable timeframe.
- **Algorithmic Limits in Large-Scale Factoring**: Classical factoring methods for large integers are limited by arithmetic and sophisticated mathematical complexity. For example, factoring a 2048-bit number is a task that would be extremely difficult for classical computers, whereas advanced quantum algorithms are designed to address this.
- **Lack of Access to Quantum Properties**: Classical computers are restricted to the "lone 1s and 0s" of bits. They do not have access to properties like **superposition** and **entanglement**, which provide "many dimensions of working memory" that are unavailable to classical systems.
- **Addressing the Limits of Moore's Law:** We can no longer rely on transistors getting smaller and smaller to increase performance. As we approach the atomic level, classical physics fails and quantum effects come into picture, necessitating a new architecture based on quantum mechanics to continue making progress.

DIFFERENCES BETWEEN CLASSICAL AND QUANTUM COMPUTERS

The primary differences between classical and quantum computers centre on their fundamental units of information, their underlying physics, and how they handle complex problems.

Fundamental Differences

- **Basic Unit of Information (Bit vs. Qubit)**:
    - **Classical computers** use bits, which are "lone 1s and 0s".
    - **Quantum computers** use quantum bits, or **qubits**, which can exist in a **superposition** represented by a linear combination of both 0 and 1 states simultaneously.
- **Information States**:
    - **Classical bits** are limited to two discrete states.
    - **Qubits** can exist in infinite states within a complex Hilbert space until they are measured.
- **Computational Principles**:
    - **Classical computing** relies on deterministic logic circuits made of hundreds or thousands of gates (like AND, OR, NOT) to perform arithmetic operations.
    - **Quantum computing** utilizes quantum mechanical properties such as **superposition** and **entanglement** to provide many dimensions of working memory that are unavailable to classical systems.
- **The Role of Measurement**:
    - **Classical data** can be copied freely from memory into a working register for use.
    - **Quantum data** cannot be copied due to the **No Cloning Theorem**. Additionally, the act of measuring a qubit forces it to "collapse" irreversibly from its superposition into a single classical state (0 or 1).

Performance and Problem Solving

- **Scaling and Complexity**:
    - **Classical computers** often face "exponential growth" in resource requirements (time or memory) as a problem size increases, making certain tasks impossible.
    - **Quantum computers** can potentially turn these exponential-time problems into manageable **polynomial-time** problems.
- **Algorithmic Efficiency**:
    - **Classical factoring** for large integers is limited to sub-exponential time methods at best; there are no known classical polynomial-time algorithms for this task.

    - **Quantum algorithms**, such as Shor's algorithm, are designed to solve these same problems with significantly higher efficiency.
- **Randomness**:
    - **Classical algorithms** are typically deterministic, meaning they always produce the same answer for the same input without inherent randomness.
    - **Quantum algorithms** are non-deterministic and involve probability and choices that can result in different internal paths to an answer.

Physical and Structural Differences

- **Hardware Architecture**:
    - **Classical hardware** is general-purpose and used for virtually all modern digital tasks.
    - **Quantum hardware** is still nascent and often requires specialized environments to maintain **coherence**; as of 2019, hundreds of physical qubits were needed to create just one stable "logical qubit".
- **Data Replication vs. Teleportation**:
    - While classical systems rely on **data replication**, quantum systems may eventually rely on **teleportation** to move information because qubits cannot be copied.

Summary of differences:

Feature	Classical Computer	Quantum Computer
Basic Unit	Bit (0 or 1)	Qubit (Superposition of 0 and 1) +3
Logic	Boolean Logic	Quantum Gates (Superposition, Entanglement)
Duplication	Can copy data	No Cloning Theorem (cannot copy)
Output	Deterministic	Probabilistic (collapses on measurement)
Best For	Everyday tasks, small problems	Chemistry, cryptography, large-scale optimization

QUBIT: THE BASIC UNIT OF QUANTUM COMPUTING

Definition: A **qubit**, or quantum bit, is the "fundamental information unit of quantum computing". While a classical bit can only be in one of two discrete states (0 or 1), a qubit is a two-state quantum mechanical system that can exist in an infinite number of states within a complex Hilbert space.

Qubit Representations

Qubits are represented through several mathematical and physical frameworks:

- **Ket Notation (Dirac Notation)**: The state of a qubit is typically written as $|\psi>$. The standard basis states are |0> and |1>, which are the quantum analogs of the classical 0 and 1.
- **Vector Representation**: In a two-dimensional complex vector space, qubits are represented as column vectors:
  - $|0> = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$
  - $|1> = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- **Linear Combination (Superposition)**: A general qubit state is represented as $|\psi> = a|0> + b|1>$, where a and b are complex numbers called "probability". The sum of the squares of their absolute values must equal 1 ($|a|^2 + |b|^2 = 1$).

Types of Qubit Bases

- **Computational Basis (Z-basis)**: Consists of the states $|0>$ and $|1>$.
- **Hadamard Basis (X-basis)**: Consists of the states |+> and |->, defined as:
  - $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
  - $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$
- **Circular Basis (Y-basis)**: Consists of the states $|i\rangle$ and $|-i\rangle$, involving imaginary numbers:
  - $|i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$
  - $|-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$

- **Logical vs. Physical Qubits**: In the context of hardware, the qubits has been distinguished as **physical qubits** (the actual noisy quantum devices) and **logical qubits** (error-corrected qubits formed by combining many physical ones).

BLOCH SPHERE (Graphical Representation of Qubits)

A single qubit can be visualized as a point on the surface of a unit sphere in three-dimensional space, known as the Bloch sphere {see figure 1}. It maps the complex mathematical state of a qubit onto the two-dimensional surface of a sphere in three-dimensional real space (R^3).

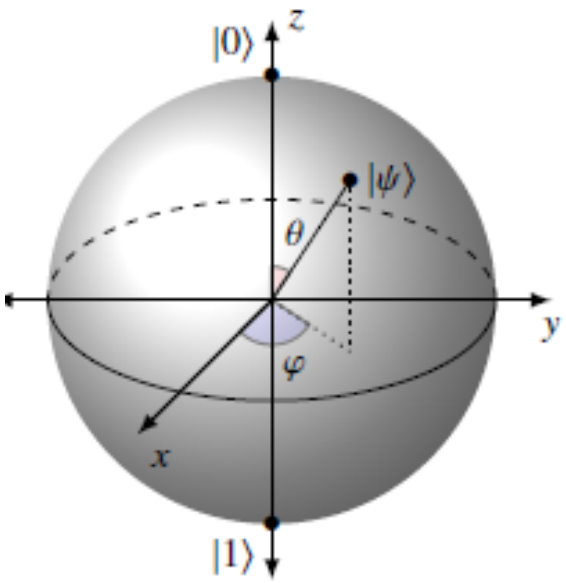


Figure 1: Bloch sphere representation of a single qubit

Core Characteristics

- **Mathematical Foundation**: A qubit's state is described by the vector as $|\psi >= a|0> + b|1>,$. Through a non-linear projection and by ignoring the "global phase" (which is not physically observable), this state can be represented using two angles: θ and ϕ
- **Coordinate Representation**: The state on the sphere is defined by:

  - $|\psi\rangle = \left\{\cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}\right\}$

    where, $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq 2\pi$

- **Visualizing States** :
    - θ: Measured from the positive z-axis.
    - ϕ: Measured from the positive x-axis in the xy-plane {see figure 1}.

Key Points on the Sphere

The sphere provides a visual map for standard quantum bases:

- **Z-axis (Computational Basis)**: The "North Pole" represents the state $|0\rangle$ $(\theta = 0°)$, and the "South Pole" represents $|1\rangle$ $(\theta = 180°)$
- **X-axis (Hadamard Basis)**: The points where the x-axis intersects the equator represent the $|+\rangle$ and $|-\rangle$ states.
- **Y-axis (Circular Basis)**: The points where the y-axis intersects the equator represent the $|i\rangle$ and $|-i\rangle$ states.
- **The Equator**: Any state located on the equator of the sphere has an equal probability (0.5) of resulting in $|0\rangle$ or $|1\rangle$ when measured.

Bloch Sphere Representation of Qubits

The Bloch Sphere visualization, named after physicist Felix Bloch, is a geometrical representation of a qubit's state space. It is a unit sphere, which is a sphere with a radius of 1. The vertical z-axis is labelled with |0⟩ at the top and |1⟩ at the bottom. The ends of two perpendicular horizontal axes, the x and the y, are labelled |+⟩ and |-⟩, and |i⟩ and |-i⟩, respectively.

The z-axis represents the probability of the qubit being measured as a 0 or a 1.

The x-axis represents the real part of the state vector.

The y-axis represents the imaginary part of the state vector.

A qubit is a combination of two states, |0⟩ and |1⟩. While in superposition, each has a complex number coefficient. With α and β representing the coefficients, the qubit's state |ψ⟩ is written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \qquad \text{------- (1)}$$

The angles θ (polar angle) and ϕ (azimuthal angle) of the Bloch sphere are calculated from these complex numbers. The angle θ is the angle between the state vector and the z-axis, while the angle ϕ is the angle between the state vector counter-clockwise from the positive x-axis. A third calculation is the length of the vector, which is 1 for pure states and less than 1 for mixed states.

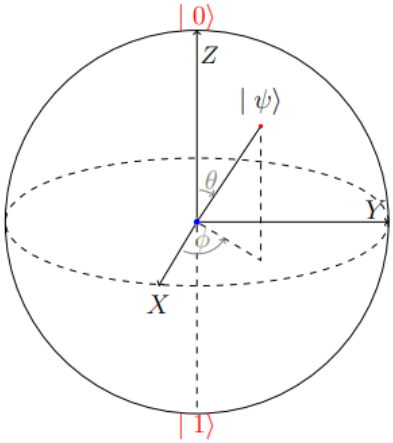
With these three values, the previous equation for the qubit can be rewritten as:

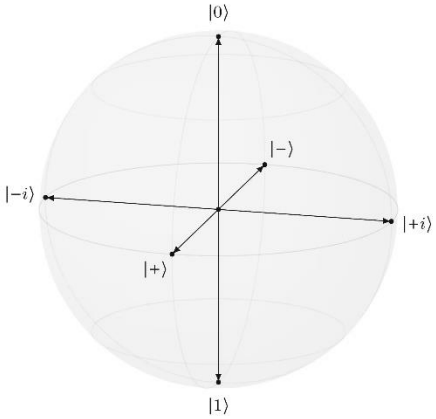
$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\phi}\sin(\theta/2)|1\rangle$$

-------- (2)

where θ ∈[0, π] and ϕ ∈ [0, 2π)

A Bloch sphere visualizes the relationship between a quantum state and two important angles, θ and ϕ. In digital mode quantum computing, quantum states are transformed by providing these angles of rotation.





Qubit Operations and Quantum Gates

When using qubits or bits, we need to manipulate them in order to turn the inputs we have into the outputs we need.
Quantum gates are the fundamental, reversible building blocks of quantum circuits that manipulate qubits (quantum bits) using quantum mechanical principles like superposition and entanglement. Unlike classical logic gates, they are represented by unitary matrices and enable parallel processing of information.

Key Aspects of Quantum Gates:

Functionality: They act as logic elements in quantum algorithms, allowing qubits to exist in 0 and 1 states simultaneously.
Representation: Quantum gates are unitary matrices, acting as 2^n x 2^n complex matrices ***(U)*** for n qubits that preserve probability, ensuring $\boldsymbol{UU^{\dagger} = U^{\dagger}U = I}$ (where $\boldsymbol{\dagger}$ is the conjugate transpose). Because ***U*** is unitary, the sum of probabilities of all possible states remains 1 after a gate operation, crucial for quantum mechanics.
Reversibility: Quantum gates are inherently reversible, meaning the input can be determined from the output. The inverse of a gate is its conjugate transpose ($\boldsymbol{U^{\dagger}}$), which is also a unitary matrix.

Types of Quantum Gates:
Single-qubit gates: Operate on one qubit to flip states (e.g., Pauli-X) or create superposition (e.g., Hadamard).
Multi-qubit gates: Entangle qubits and allow them to interact (e.g., Controlled-NOT or CNOT).

Single-qubit gates

- **The X,Y & Z-Gates**
- **The H-Gate**
- **The Phase Shift Gate**

- **The X,Y & Z-Gates**

The X gate is a π rotation around the x axis of the Bloch sphere. Applying the X gate to |0⟩ results in |1⟩, and applying the X gate to |1⟩ results in |0⟩, so it is an operation similar to the classical NOT gate, and is also known as bit flip. The matrix representation of the X gate is below.

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$X|0> = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Similarly the Y & Z-gates perform rotations of π around their respective axes. Their matrix representations are

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- **The H-Gate**

The Hadamard (H) gate is a π rotation around an axis halfway between the x and z axes on the Bloch sphere. Applying the H gate to |0⟩ creates the superposition state $\frac{|0>+|1>}{\sqrt{2}}$.
The matrix representation of the H gate is below.

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

The truth table of the Hadamard gate is

Input	Output
$\|0\rangle$	$\frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle)$
$\|1\rangle$	$\frac{1}{\sqrt{2}}(\|0\rangle - \|1\rangle)$

- **The Phase Shift Gate**

The phase shift gate (R_ϕ) change the phase of the state |1⟩. Its general representation is given as

$$R_\phi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}$$

The inverse phase shift gate () is represented as

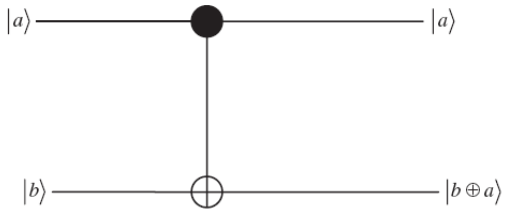
$$R_\phi^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\phi} \end{pmatrix}$$

Unlike the previous gates, the phase shift gate is not necessarily its own inverse, to reverse the change we must apply a phase shift in the other direction. Theoretically we can choose any phase shift. We call the phase shift by $\phi = \pi/4$ the T-gate and we call the phase shift by $\phi = \pi$ the Z-gate.

MULTI-QUBIT GATE

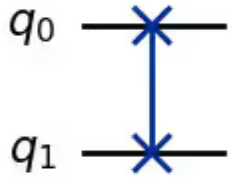
1. **CNOT (Controlled-NOT) gate:** The two input qubits are called the "control" and "target" qubits. The control qubit remains unchanged, but its state dictates what happens to the target qubit. If the control qubit is in the state |1⟩, then an X gate is applied to that target; if the state of the control qubit is |0⟩, then no change is made. In the notation below, assume the qubit A (rightmost qubit) is the control, and qubit B (the leftmost qubit) is the target. Below, the notation used is CNOT|BA⟩.

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$



Input	Output
$\|00\rangle$	$\|00\rangle$
$\|01\rangle$	$\|11\rangle$
$\|10\rangle$	$\|10\rangle$
$\|11\rangle$	$\|01\rangle$

2. **SWAP gate:** This gate swaps the states of two qubits.



$$SWAP = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Input	Output
$\|00\rangle$	$\|00\rangle$
$\|01\rangle$	$\|10\rangle$
$\|10\rangle$	$\|01\rangle$
$\|11\rangle$	$\|11\rangle$

Quantum Entanglement and Bell States

Despite all its apparent strangeness, quantum mechanics provides a remarkably accurate description of nature at microscopic scales. Its predictive power and experimental success have firmly established it as the foundation of modern physics. Beyond its theoretical importance, quantum mechanics has directly enabled a vast range of transformative technologies that shape contemporary society, such as lasers, atomic clocks, and the entire field of electronics, including computers, the internet, mobile communications, GPS, and many more. These technologies are referred to as the first generation of quantum technology because they are based on the basic principles of quantum mechanics.

These achievements naturally raise a deeper question: if the basic principles of quantum mechanics have already produced such revolutionary tools, what additional possibilities might emerge from its more subtle and uniquely quantum features? This question has motivated the development of what is now called **quantum information science**, where distinctly quantum phenomena are treated not merely as curiosities but as operational resources.

Among these phenomena, quantum entanglement occupies a central role. Unlike classical correlations, entanglement reflects intrinsically quantum relationships between physical systems, revealing aspects of reality that have no classical counterpart. It is this feature that underlies many of the conceptual and technological advances associated with modern quantum technologies.

It is often said that entanglement is a resource in quantum computation and information. One of the most unusual and fascinating aspects of quantum mechanics is that particles or systems can become entangled. For the simplest case of two quantum systems, denoted as A and B, if these systems are entangled, it means that the values of certain properties of system A are correlated with the values that those properties will take for system B. These properties can remain correlated even when the two systems are spatially separated, leading to what is often called spooky action at a distance.

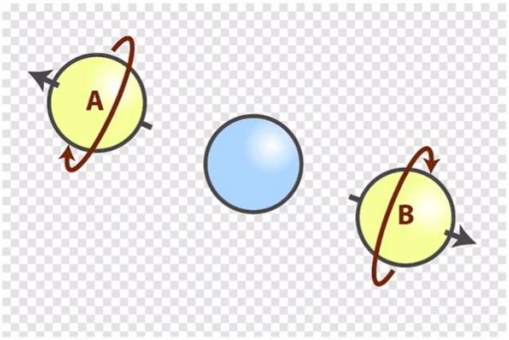


Fig. 1: Entangled particles [1]

In fact, to understand this idea more clearly, one may discuss spin quantum systems to illustrate and explain quantum entanglement. The phenomenon of entanglement can also be illustrated with a famous thought experiment known as Schrödinger's cat. In this experiment, a cat is confined in a box. A photon is incident on a 50–50 beam splitter, which means the photon can take either path. Path A or path B with equal probability. If the photon takes path A, it triggers a pistol and kills the cat, making it dead. On the other hand, if path A is empty, that means the photon has taken path B. In that case, the pistol cannot be triggered, and the cat remains alive.

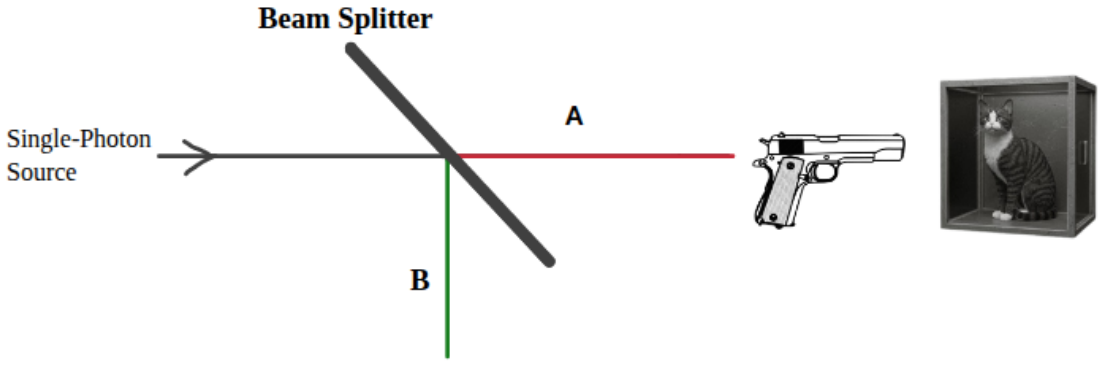


Fig.2 Diagram to demonstrate Quantum Entanglement

This situation can be described mathematically by a superposition state. If you do not open the box, the cat is both alive and dead at the same time. This is the main idea expressed by the equation. In physical terms, the cat has two possible states: alive and dead. The same is true for the photon in path A; there can either be zero photons or one photon. The important point is that the state of the cat and the state of the photon are not independent of each other; they are connected.

The cat is alive, and the photon does not go through path A, or the cat is dead, and the photon goes through path A. This means that the cat and the photon are entangled. An observer who cannot see inside the box has no idea about the state of the cat or the state of the photon. The observer also does not know whether the photon will go through the path. After five minutes or even after a hundred years, because at any given time there is only a probability that the photon has passed through path A. It may go through path A, or it may not. Therefore, the observer can only describe the total state of the system, which includes both the cat and the photon, as an entangled state. Here, the cat and the photon are two subsystems. Together, the cat and the photon form one total system. These subsystems are inseparably linked to each other.

This is because we can use entanglement to accomplish communication and information processing tasks that would otherwise not be possible. The first of these, known as quantum teleportation, is a procedure that allows one party, usually called Alice, to send a quantum state to her friend Bob without transmitting the state in the usual physical sense. By using entanglement, Alice and Bob can create a quantum communication channel that links them in a special way, often explained through the EPR paradox. This allows Alice to transfer her quantum state to Bob in a way that seems almost magical. However, faster-than-light communication is not possible through teleportation because Alice and Bob must still share a classical communication link for the process to work.

One of the most unusual and fascinating aspects of quantum mechanics is that particles or systems can become entangled. For the simplest case of two quantum systems, denoted as A and B, if these systems are entangled, it means that the values of certain properties of system A are correlated with the values that those properties will take for system B. These correlations can exist even when the two systems are spatially separated, which leads to what is often called “spooky action at a distance.”

The roots of this idea go back to the year 1935, when Albert Einstein and his two colleagues, Boris Podolsky and Nathan Rosen, now commonly known together as EPR, published a famous paper titled “Can the quantum-mechanical description of reality be considered complete?” In this paper, which was written by Einstein as a critic of quantum mechanics, they tried to show that quantum theory is incomplete and leads to unreasonable predictions. A central belief of EPR and other realists was that physical systems have definite properties with fixed values, whether we observe them or not. In other words, they believed in an objective reality. According to this view, a property of a system has a clearly defined value even before any measurement is made.

To explain EPR kind of experiment from the beginning, the so called hidden local variable theory was proposed. This went on for many years without any conclusion. However, in 1964, John Bell proposed an inequality that shows that the classical view in the form of a theory of hidden local variables satisfies certain conditions that are violated by quantum physics so hidden variable theory has to be abandoned.

Bell States:

When two systems are entangled, the state of each composite system can only be described with reference to the other state. If two states are not entangled, we say that they are a product state or separable. One simple test that can be applied to states is the following. This state is separable if and only if $ad = bc$

The members of the Bell basis, *EPR states*, $|\psi\rangle = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$ sometimes called the *Bell states* or the

$$|\beta_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \qquad |\beta_{10}\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

$$|\beta_{01}\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}} \qquad |\beta_{11}\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

Are the Bell states given above entangled?

$$|\beta_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad |\beta_{01}\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

$$|\beta_{10}\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}, \quad |\beta_{11}\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

For $|\beta_{00}\rangle$, we have a = d = $\frac{1}{\sqrt{2}}$, b=c=0, so ad= $1/2 \neq$bc. So, $|\beta_{00}\rangle$ is not a product state and must be entangled. For $|\beta_{01}\rangle$, a=d=0, b=c=$\frac{1}{\sqrt{2}}$, we conclude that $|\beta_{01}\rangle$ is also entangled.

Below we present the circuit diagram to prepare quantum entanglement using gates,

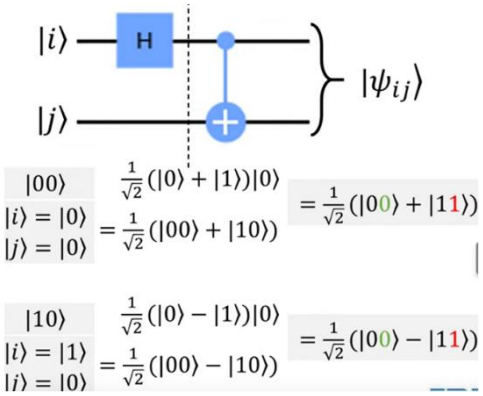


Fig.3 Quantum Entanglment circuit

Creating Entanglement Using Hadamard and CNOT Gate

Entanglement is a uniquely quantum mechanical phenomenon in which the state of one qubit becomes inseparably correlated with the state of another. A standard way to generate entanglement between two qubits is by combining a Hadamard (H) gate with a controlled-NOT (CNOT) gate.
We begin with two qubits initialized in the computational basis: $|i\rangle = |0\rangle$, $|j\rangle = |0\rangle$
The initial two-qubit state of the system is $|00\rangle$. This is a separable state (no entanglement). A Hadamard gate is applied to the first qubit. The Hadamard creates a superposition:
$H|0\rangle = (1/\sqrt{2})(|0\rangle + |1\rangle)$. The two-qubit state becomes $(1/\sqrt{2})(|0\rangle + |1\rangle) \otimes |0\rangle$. Expanding: $(1/\sqrt{2})(|00\rangle + |10\rangle)$
At this stage, superposition exists, but the state is still not entangled. Next, a CNOT gate is applied with the first qubit as control and the second as target.
CNOT action:
- If control = 0 → no change
- If control = 1 → flip target qubit
Applying CNOT to each term: $|00\rangle \rightarrow |00\rangle$, $|10\rangle \rightarrow |11\rangle$. Final state: $(1/\sqrt{2})(|00\rangle + |11\rangle)$

The final state is one of the Bell states, representing maximal entanglement. It cannot be written as a product of single-qubit states. Measuring the first qubit immediately determines the state of the second qubit.
Example: If the first qubit is measured as 0, the second is 0. If the first qubit is measured as 1, the second is 1.

References:

[1] NPTEL IIT Guwahati. n.d. “Quantum Entanglement: Fundamentals, Measures, and Applications.” YouTube video. https://www.youtube.com/watch?v=xifUo79u4eE.

[2] McMahon, David. 2008. *Quantum Computing Explained*. Hoboken, NJ: John Wiley & Sons.

Sample Questions:

1. For two quantum systems, A and B, what does it mean for the systems to be entangled?
2. Using the Schrödinger's cat thought experiment, explain how the photon and the cat become entangled.
3. What is meant by a separable (product) state? Using the condition ad=bc, explain how one can determine whether a two-qubit state is entangled.
4. Starting from the initial state $|00\rangle$, describe step-by-step how applying a Hadamard gate followed by a CNOT gate generates an entangled Bell state.
5. Perform the same steps mentioned in 4. for the states $|01\rangle$, $|10\rangle$, $|11\rangle$.